History dependence without qualified common support: Limits of learned native bitrate control

networkRLCD project
Retrospective research-decision synthesis

15 October 2026 — negative-result edition, evidence through PO1

Abstract

Fine-grained learned bitrate control requires a defensible connection between sender information, randomized action effects and future delivery risk. We synthesize frozen native Chromium and simulator studies that did not establish that connection robustly. A local native study found a positive secondary BWE-by- action PSNR interaction, but a direct threshold-policy trial reduced timely delivery by 14.1 percentage points (95% interval $[-22.6, -5.5]$) relative to fixed 0.65. An independently sourced replication reversed the descriptive quality interaction in both measured campaigns but failed registered cell coverage, so it supplies no population reversal estimate. Subsequent native support studies failed occupancy or repeatability before their action stages. A qualified randomized-prefix experiment nevertheless establishes finite-horizon causal history effects on native BWE, with content/time heterogeneity and no registered sustained convergence by 36 s. Matching, conditioning and phase-aware rules did not produce prospectively qualified common state. Available mechanism channels did not explain residual divergence under their frozen criteria; actual internal transport timing could not be exposed with both causal freshness and stable identity. A proposed instrumented browser was never built because resource admission failed. Finally, a separately preregistered 816-episode simulation screen failed out-of-family prediction, action-discrimination and useful uncertainty gates against strong simple baselines. Its native panel remains unopened. We close the current fine-grained learned-control objective under these fixture/data assumptions. This is neither a universal impossibility result nor evidence that all history is irrelevant. We specify the distinct evidence needed for a new fixture or a different episode-level estimand, without selecting a successor or fitting another controller.

1 Research question and contribution

Interactive media must balance source quality, timely delivery and network cost; these goals are not interchangeable [1]. A native sender's reported bandwidth estimate (BWE) is an output of a transport/encoder feedback system, not an exogenous measurement of usable capacity. A controller could exploit observable history only if its action-value and risk claims are identifiable for the relevant population and survive validation. Good mean prediction in one regime is insufficient; useful uncertainty must also transport [2].

The original networkRLCD program investigated calibrated learned adaptation in simulation, then native action effects and causal support. This edition changes the central paper claim to the accumulated negative finding. It does not erase earlier positive or negative simulator studies, replace their results, or claim their mechanisms are native WebRTC. Its evidence anchor is commit `f109ab2`, the completed PO1 result. All prior tracked artifacts, including the earlier controller-centric manuscript, are preserved.

The contributions are: (i) a separation of bounded native action evidence from prospective support failures; (ii) qualified causal history dependence alongside unsuccessful attempts to obtain useful common state; (iii) a precise distinction between observability limits and an unattempted instrumented build; and (iv) a bounded synthetic negative result that requires calibration and supported action discrimination as well as accuracy. We report a research stopping decision, not a new bitrate controller or safety certificate.

Earlier simulator studies retain their reported local successes and failures in the frozen prior edition [5]. They used different representations, policies and targets; PO1 does not erase those results or supply their native validation. Passing some synthetic population-panel criteria did not establish uniform within-family calibration or native control value.

2 Estimands and what failure means

Let $\mathcal{I}_t$ contain sender observations, past commands and records *actually received* by decision time t. Let U_t denote unobserved transport state, A_t a bitrate action and Y a specified future endpoint. Three questions must remain separate:

$$\tau_H(t; S) = \mathbb{E}[\log_2 B_t(H_d, S) - \log_2 B_t(H_a, S)], \tag{1}$$

$$\tau_A(s; \mu) = \mathbb{E}_\mu[Y(a_1) - Y(a_0) \mid g(\mathcal{I}_t) = s], \tag{2}$$

$$V_\mu(\pi) - V_\mu(\pi_0) = \mathbb{E}_\mu[Y(\pi) - Y(\pi_0)]. \tag{3}$$

The first is a total effect of randomized early service order H followed by common input service S. The second is a conditional action effect under a specified history/episode distribution μ. The third is a complete policy or sequence contrast. A factual predictor $\mathbb{E}[B_{t+1:t+h} \mid \mathcal{I}_t, \text{logging regime}]$ is a fourth, predictive target; it is not a native counterfactual action value.

Randomized actions can identify local or marginal effects despite hidden U_t, subject to consistency, actual randomization/positivity and the stated population. Observing every GCC mechanism or finding a stationary Markov state is not a universal prerequisite for randomized policy inference. Native V9 and V10 demonstrate that bounded causal contrasts can be estimated in this fixture. The project's intended *transferable fine-grained conditional value*, however, additionally needs support and a defensible relationship between the selected observable context and its future response across histories and sources. That relationship has not qualified. Changing μ can change the latent-history mixture even when instantaneous BWE is the same.

We distinguish a contradicted specific superiority claim, an unsuccessful registered operational recipe, an unestimable comparison after a gate failure, and a fixture/practical limit. None alone is a theorem that all learned native control is impossible. Failed support sessions are not treatment outcomes; zero emitted predictions have unestimable precision. Receiver outcomes left unopened in support studies are not assigned zero effects.

3 Evidence handling, sources and technical controls

This is a synthesis of committed result reports and their pinned provenance, not a meta-analysis or raw-data reanalysis. No captures are restored, no closed validation representation is searched, no receiver outcome is opened and no new model is fitted. The evidence ledger binds all cited reports, protocols, selected storage receipts and manuscript inputs to SHA-256. A preservation audit checks every file tracked at the evidence anchor. The new edition is separate so that old scientific freezes remain valid.

Native Chromium outcomes, native responses to controlled synthetic service, measured-rate replay and simulated outcomes are distinct. Native V9/V10 used generated content and a controlled schedule; their native telemetry is not a fluid simulation. V11D crossed licensed source content with four strata nested in two measured campaigns. V12 used three campaigns. V13's LTE/EVDO bundles come from one measurement program. M5 onward reuse two licensed clips and measured-source apparatus; many later common-service plateaus are controlled inputs. A fresh browser process is a fresh run, not a new independent content production or network campaign. Measured application-throughput replay is not a live Internet trial or an observation of every end-to-end queue.

The managed browser policy prohibits nonproxied UDP. Qualified native acquisition used TURN/TCP without changing that policy. Sender/relay clock markers, actual-run held-out clock coverage, native initialization, cap/jitter readback, replay fidelity and cleanup were enforced before admitted analyses. Later receipt-based features could not use future clock anchors or backdated event timestamps. Technical qualification is necessary for attribution, but is not scientific common support. We retain original failures and prospective repairs as separate studies rather than replacing failed originals.

4 Native action evidence did not establish transferable value

4.1 A local secondary signal and a harmful policy

Native V9 randomized0.65 versus 1.05 BWE-relative caps at six3s boundaries in 48 sessions, nested in 12 generated-scene seed clusters. Eleven clusters met the registered state/action counts; one remained in the evidence but outside the complete-cell analysis. The primary high-minus-low BWE interaction in the150ms on-time fraction was -1.8 percentage points, with an unadjusted95% t interval $[-29.4, 25.8]$. The secondary PSNR interaction was

0.508dB [0.077, 0.938] [6]. PSNR is sampled among on-time frames, not common-frame quality under all actions or human QoE. This supports a local secondary signal under that logging regime, not a stable state value or policy benefit.

Native V10 tested the direct whole-session implication: use1.05 above the frozen 500 kbps BWE boundary and 0.65 below, versus fixed 0.65. All48 sessions qualified; two repeats/policy were averaged within each of 12 scene seeds. The threshold policy's on-time contrast was −14.1 percentage points [−22.6, −5.5]. Overall PSNR was +0.257dB [−0.132, 0.646]. The fixed threshold's local superiority claim is contradicted [7]. This policy is closed and is not retuned. A positive quality interaction did not translate into whole-session delivery improvement.

4.2 Under-covered replication and action stages that never opened

V11D completed64 qualified native captures from four content groups, four network-family strata and two measured campaigns. Only9/16 crossed cells had at least two holds in every state/action bucket; at least 13 were required. Only seven cells had PSNR in every bucket. Their descriptive PSNR interaction was −0.609dB, with campaign means −0.419dB (NordicDat) and −0.862dB (Rome) [8]. Both directions oppose V9, but the coverage gate failed. We call this a negative, under-covered replication, not an estimate of a population reversal. No missing bucket is imputed or used to manufacture an independent campaign sample.

V12 attempted all 24 support originals before randomizing any action. Twenty-three ordinary captures sealed and one preserved capture failed native media statistics. Ten passed all gates; none of six cells had complete frozen low/high-BWE support. The23 ordinary captures passed clocks, transport, replay and cleanup. All48 planned action sessions stayed unrun [9]. Capacity-based source hints did not guarantee native estimator occupancy.

V13 replaced the binary split with a preregistered continuous BWE interaction, but still required support before actions. All32 originals individually qualified. Six of 16 content/profile contexts exceeded the0.25-octave fresh-repeat limit; only one of four cells passed completely. All cells passed profile range and SD, yet the cross-cell support intersection was empty. Its64 randomized action sessions remain unrun [10]. These failures provide no action-effect or QoE result. They constrain the design assumptions under which a later effect would be interpreted.

5 History is causal; usable common state did not qualify

5.1 M5's supported finite-history effect

M5 qualified 84 original sessions and 42 randomized prefix pairs before its causal comparison. Within each of two reused contents,21 adjacent pairs received ascending/descending permutations of the same early service multiset and then the same measured-source suffix. Actual-run clocks and suffix-exposure checks passed [11]. At18s, descending-minus-ascending mean log2-BWE differences were 0.20746octave [0.09910, 0.31413] for Kampar and 0.14750 [0.09807, 0.19353] for Volga. Both registered distribution tests reject at local $\alpha = 0.025$.

Later signed effects changed direction and neither content satisfied sustained convergence through 36 s. Thus early service history causally changes later native BWE in this fixed recipe. This does not establish permanent memory, behavior beyond the horizon, independent network replication, or a unique mediator. Holding later service input common does not force identical internal queue, encoder or estimator trajectories.

5.2 Matching, conditioning and dynamic-state failures

M6R qualified 32 originals but only four of six frozen content/landmark strata met membership coverage. Adding recent encode-time history excluded zero BWE-only matches; the selected state had exactly the baseline membership. Conditional future comparisons were therefore not run [12]. This is sparse overlap and absent matching discrimination at those calipers, not a null causal history effect. Matching on post-prefix BWE/history also cannot by itself identify a direct prefix effect or causal mediation.

M7RA's four feasibility originals passed technical gates, but only two reached the instantaneous target and one met sustained-history membership. Neither region qualified across histories. Its128-session validation remains untouched [13]. A fixed conditioning recipe did not create common state; similar census means could not replace sustained occupancy.

M8's16 characterization originals under repeated24s controlled-service plateaus showed strongly asymmetric native dynamics: rise t50 about 7.4–10.2 s and t90 13.3–15.9 s, versus fall t50 about 0.39–0.79 s. At each frozen candidate age, none of 64 nested plateau occurrences passed future verification under the D1 local-settling rule.

No conditioning age was selected [14]. These are native responses of content, encoder and transport to controlled service, not isolated GCC parameters. The onset search was censored and cannot prove permanent nonsettling.

M9R prospectively extended observation to a64s common plateau; all eight instrumentation and 64 validation originals qualified. Instantaneous BWE (B0) tagged 64 originals and D1 tagged 55, with no future-verified first predictions. The phase-aware P2 rule emitted no tag, making its positive precision unestimable. Full four-pair matching coverage fell from42/64 nested strata for BWE-only to13/64 for the compact history vector, and residual future divergence remained in some matched contexts [15]. Later observation did not establish prospectively useful state. It licenses neither another fixed wait nor a classifier on these closed captures.

6 Mechanism observability and build practicality

6.1 M10R: available counters did not resolve residual divergence

M10R qualified four instrumentation and eight fresh development originals at 1.414 Mbps common input. It observed acknowledged owned-relay queue/service counters, native RTCP RTT, RTT-counter detection age, send/cap history and encoder behavior. No channel met the frozen repeated same-direction contrast with controls in both contents. Of96 nested contrasts,49 compact-matched and 23 exceeded the0.25-octave future-distance bound. There were 0/400 nested positive stationary labels, so no stability classifier was fitted [16].

RTT-counter detection age is not transport-CC generation/arrival age; the owned relay is one queue, not every queue in the TURN/TCP path. Available contrasts may be consequences of BWE or shared causes. This assay established neither causal mediation nor a stationary or prospectively reproducible trajectory state. Some repeatable moving marginal distributions are insufficient.

6.2 M11R: real events lacked jointly useful freshness and identity

Both M11R native capability originals qualified ordinary acquisition. Identified RTC logs contained actual TCC, delay/loss estimator-update and probe events. The gate required at least 90% of each family sender-available within 250 ms, clock uncertainty at most50ms and stable PC/SSRC/controller attribution. TCC freshness was1.78%/1.89%; BWE-update freshness1.42%/1.77%; probe freshness 0/16 and 0/31. First identified export was about 59 s after measurement start, followed by roughly5s batches [17].

Verbose target-update lines were fresh but lacked controller identity; a target emission is not every estimator update. Trace exports supplied no online sender receipt. No relay/socket/RTT/RTP-burst equivalent proxy qualified. Thus all required channels qualified in 0/2 originals. Native transport-timing explanations remain untested in this fixture, not disproved. A record becomes causally usable at its receipt, not its earlier producer timestamp.

6.3 IF1: the instrumented alternative was not built

IF1 specified pinned Chromium/WebRTC hooks, stable identity, nonblocking event publication and sender-clock receipt. Its formal admission audit found17.7 GB free disk on a33.8 GB filesystem, below the source's100GB minimum and IF1's 130 GB/16GiB/4-core build requirement; no qualified remote builder or artifact was available. There were zero compiler attempts, C++ instrumentation patches, full source checkouts or new native sessions [18]. This is a practical fixture-admission failure. Neither live instrumentability nor instrumented Chromium's scientific informativeness was tested. Static hook discovery is not channel qualification, common support or action value.

7 PO1: three distinct synthetic failures

PO1 was preregistered as a cheap necessary-condition screen before a fresh native factual panel. It uses the existing fluid bottleneck, simplified AckedBwe and assumed encoder response, not production libwebrtc [4, 19]. Three fits use384 fitting,144 calibration and 288 held-out validation episodes in total; validation covers four generator families disjoint from fitting/calibration. All816 episodes and all binary training/calibration support gates qualified. Models/calibrators were locked before validation decoding. Six decisions per episode are nested; uncertainty resamples episodes within fit, not holds.

The fixed 1.05 synthetic logger is distinct from the frozen native fixed 0.65 baseline. At each decision, the feature vector is saved before hindsight branches compare0.65 for5s with 1.05 for3s then0.65 for2s. A separate

factual branch continues logging1.05 for5s. Only the simulator supplies those action branches; none is transferred to native data.

S1 uses current values/masks and past3s means, slopes and missing fractions of BWE, delivered throughput, RTT, requested cap and actual send rate. H1 adds only ordered1/2/3s samples/masks. These are fixed-penalty linear/logistic models; there is no recurrent model, state search or learned policy. Baselines include persistence, trailing mean, clipped trend, train-only constant and instantaneous linear models as well as S1. Calibration is held out; proper prediction scores are not themselves control guarantees [3, 2].

	Factual BWE MAE (octaves)			Synthetic advantage MAE		
Family	Persistence	S1	H1	Constant	S1	H1
sawtooth	0.740	0.457	0.469	0.256	0.204	0.203
square_wave	0.684	0.413	0.421	0.251	0.200	0.199
slow_fade	0.587	0.447	0.469	0.183	0.152	0.154
feedback_flap	0.407	0.716	0.724	0.080	0.221	0.225

Table 1: PO1 frozen validation values: 72 episodes/family across three fixed fits, six nested decisions/episode. S1 is the fixed-history summary baseline; H1 adds ordered lags. Lower is better. These are simulator results, not native effects.

Prediction did not transport robustly. H1 improves on persistence/current-value baselines in three families, but its BWE MAE is worse than S1 in every family. Under feedback-flap H1 error is0.724 versus persistence0.407,77.8% worse; S1 also fails there. This is evidence against this package's robust out-of-family incremental information, not a claim that all history is irrelevant. The assay tests a representation/model/logging/calibration package, not the information-theoretic sufficiency of every possible function of the history. That limit does not waive the registered stop on complexity escalation.

Action discrimination did not qualify. Advantage-MAE gains of H1 over S1 were only 0.19% and 0.43% in two families, below the frozen5% requirement, with lower bootstrap improvement bounds below zero and fit-level sign reversals. The other families worsened. Feedback-flap H1 advantage MAE0.225 versus constant0.080 is179.4% worse. These are synthetic action-value prediction errors, not native causal effects or an executed policy's reward. Local oracle headroom does not ensure that causal observable history can discriminate actions under shift.

Coverage, calibration and supported uncertainty also failed. Nominal90% sequence intervals use calibration-episode maximum absolute residuals. H1 BWE full widths are 4.08–5.40 octaves versus maximum 1.0; advantage widths2.96–3.31 utility units versus maximum 0.50. Aggregate coverage passed its minimum and within-fit coverage was at least 75%, but excessive widths fail usefulness. This empirical cross-family calibration has no distribution-free OOD guarantee. Every family fails the binary calibration conjunction; feedback-flap ECE is0.397 for BWE drop and 0.321 for beneficial advantage. That family's zero unsafe-positive labels cannot establish risk sensitivity.

An offline supported-action assertion requires a positive advantage lower bound, benefit probability at least 0.90 and unsafe probability at most0.10, with enough distinct episodes to assess errors. None qualified in any family, versus a minimum 30 asserted episodes/family. Zero assertions are insufficient supported discrimination, not demonstrated safety. No learned action was executed. The registered stop leaves all 64 native factual sessions untouched; there is no PO1 native predictive result [19].

8 Research decision and bounded alternatives

The active fine-grained learned native-control objective is closed under the available fixture/data assumptions. No state passed its prospective support contract; the mechanism route reached an observability boundary; the new fixture was not admitted; PO1 did not supply robust predictive discrimination or useful uncertainty. These failures cannot be repaired by pooling sources, relaxing calipers, opening unused allocations, extending waits, increasing model complexity or searching closed validation. The PO1 stop remains exact: no history expansion, retraining with more complex sequence models, calibration/threshold tuning, native factual panel, controller fit or safe-policy efficacy work from that representation.

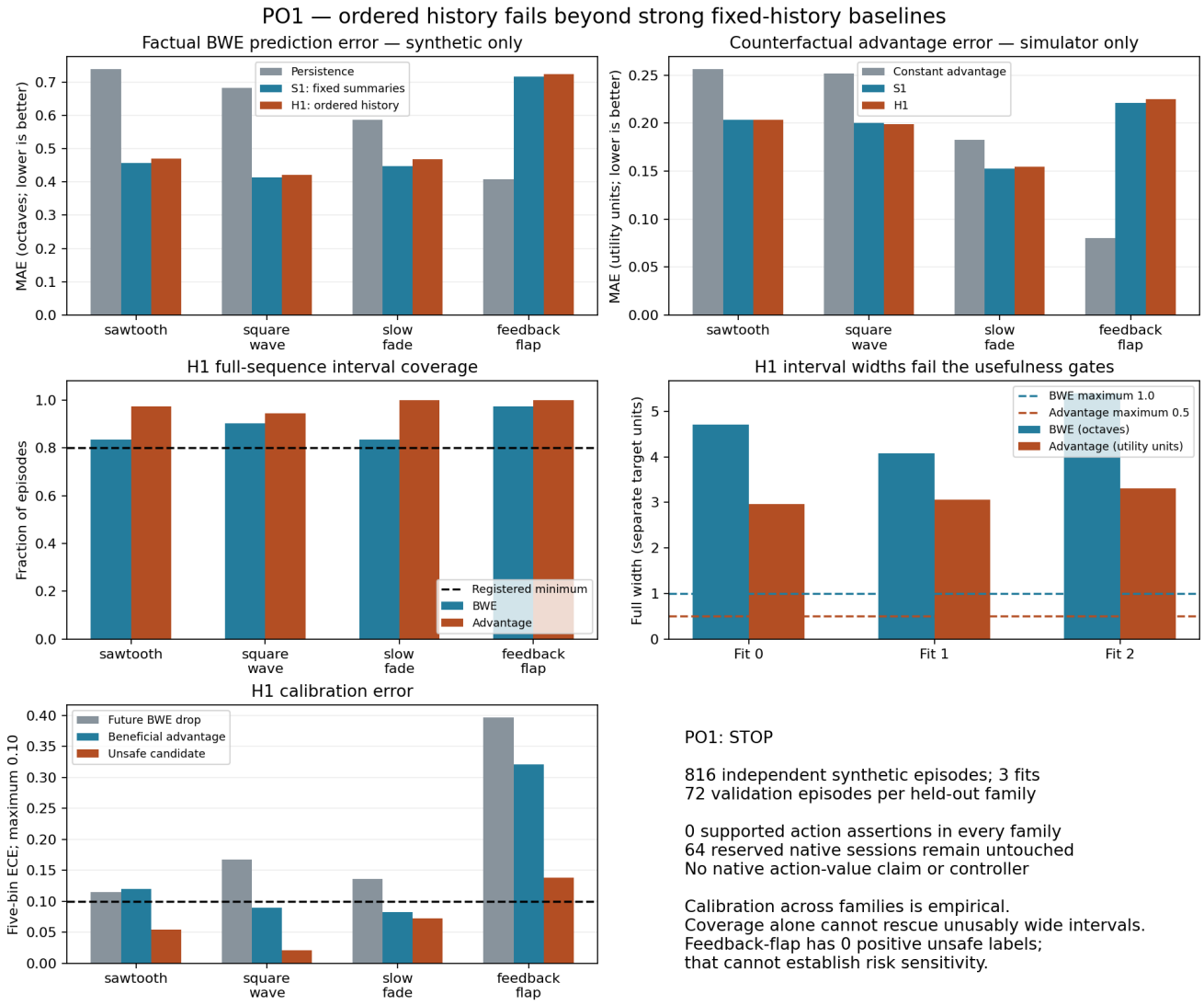


Figure 1: Frozen PO1 metrics from its committed report, with no new analysis. The baseline comparisons, calibration error and interval widths must be read alongside coverage. All families fail the preregistered conjunction. Families, fits and decisions are not independent simulator implementations or native replications.

We identify only two conditional changes in premise, neither selected:

A new admitted instrumented fixture. An attested builder or verifiable artifact must first make IF1 practical. Actual transport-feedback, estimator-update, probe and pacer hooks must qualify identity, causal receipt, freshness, timestamp uncertainty, noninterference and drop accounting in fresh runs. Fresh technical and observability panels must precede a new untouched support panel under the original applicable gates. Only then could a separately preregistered action study test conditional value and risk over strong fixed 0.65 across independent content/campaigns. Counters alone neither identify a sufficient state nor justify controller fitting. No capacity change or build is attempted in this synthesis.

A genuinely different episode/sequence estimand. A new research question could randomize complete simple prespecified sequences at independent episode boundaries under an explicit initial/history-mixture population, treating within-episode carryover as part of the intervention. This requires its own permitted fixture, causal/positivity assumptions, source/campaign clusters, strong fixed 0.65 comparator, risk/precision gates, stopping rule and fresh data. It cannot reuse PO1's representation or unopened native allocation, assert washout, or retroactively waive a state gate. Replicated episode-level benefit would establish that bounded contrast only. Before fitting an episode-level selector, a separate prospective study would still need supported sequence-value discrimination and uncertainty under new source splits. Such evidence would not automatically justify fine-grained conditional control. No sequence protocol, efficacy panel or learner is instantiated here.

The default outcome is the present negative-result paper. If neither changed premise is independently valuable and feasible, the scientifically defensible decision is to stop, rather than nominate a successor to continue searching.

9 Limitations and nonclaims

These are adaptive research questions with prospectively frozen individual studies, not one preregistered omnibus experiment. Their sample counts cannot be summed into a population confidence claim. Generated scenes, reused clips, directions/windows, repeated landmarks and controlled inputs stay within their source/campaign clusters. PSNR and 150 ms frame-readback delivery are local proxies, not human QoE or live deployment safety. Technical qualification does not establish an empty end-to-end path queue or complete estimator observability.

Specific initial instrumentation failures remain failures: M6's clock extrapolation, M7's early timers, M7R's launcher, M9's two excessive sample gaps, M10's cleanup overrun and M11's SSRC0 identity. Prospective repairs use new captures without changing numerical gates. None is recast as a scientific null or erased by a successor. IF1 was unattempted at build stage. P2 emitted no tags. Neither circumstance supplies evidence for a zero mechanism effect.

We have not proved permanent history memory, universal nonsettling, causal mediation by an internal queue, impossibility of instrumented Chromium, or a general inability to identify randomized native policy effects. Synthetic calibration failure is not native calibration failure; no PO1 native prediction was measured. Conversely, a bounded positive simulator or local native estimate does not establish broad native control value. This paper documents the missing evidence rather than replacing it with a stronger claim.

10 Conclusion

Native early-service history has a qualified finite-horizon causal effect. Local randomized action and policy contrasts can be measured, including a harmful threshold policy. Yet the registered route to reproducible native conditional value did not qualify. Current diagnostics leave internal timing unobserved for causal online use; the instrumented build was never attempted; the compact synthetic history screen failed strong baselines and useful uncertainty under family shift. Together these results justify stopping the current fine-grained learned-controller objective under the available fixture and data assumptions. All frozen evidence and unused panels remain intact. New measurement capability or a separately justified episode-level question would require new admission evidence and fresh studies; no controller is fitted or promoted by this synthesis.

A Reproducibility and provenance

The new manuscript and decision document are versioned separately from every file at`f109ab2`. The machine-readable claim/source ledger is `data/research-decision-rd1/evidence-ledger.json`; the prior-file freeze is `data/research-decision-rd1/prior-freeze-sha256.json`. The RD1 audit records summary-only input access, unchanged prior bytes and closed-panel disposition. The PDF, sources and audit are preserved in a private immutable archive with fresh object/member size and SHA-256 verification. No licensed media, raw native session, new receiver endpoint or learned policy is created by RD1. This retrospective synthesis does not register a new experiment.

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[5] networkRLCD. RLCD: Calibrated Reinforcement Learning for Safe Real-Time Media Adaptation. Earlier frozen working manuscript at commit `f109ab2`; `paper/main.tex`. Historical simulator results retain their original scope; this negative-result edition supplies the current decision.

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[9] networkRLCD. V12 execution: support qualification failed. Frozen report. `docs/NATIVE_SUPPORT_VALIDATION_V12_RESULTS.md`.

[10] networkRLCD. V13 execution: support failed; action study closed. `docs/NATIVE_CONTINUOUS_VALIDATION_V13_RESULTS.md`.

[11] networkRLCD. M5: qualified instrumentation makes the prefix-history effect estimable. `docs/NATIVE_PREFIX_MEMORY_M5_RESULTS.md`.

[12] networkRLCD. M6R: history-state common support did not qualify. `docs/NATIVE_HISTORY_STATE_M6R_RESULTS.md`.

[13] networkRLCD. M7RA: qualified instrumentation, failed prospective state creation. `docs/NATIVE_STATE_CONDITIONING_M7RA_RESULTS.md`.

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